MATH INQUIRERS HOLIDAY PROGRAM

Free 30-minute taster

Lesson 1: Million Puzzles (Numbers to 10 Million) • Math Inquirers P5 Ages 10-12

Notice it. Test it. Explain why.

What this is

A short sample of a real Math Inquirers lesson for Ages 10-12 (finishing P5). It is Lesson 1 of the Primary 5 holiday program (Million Puzzles), shortened to about 30 minutes: a warm-up, a hands-on explore, a mini-lesson, a few worksheet questions and an exit ticket. The same routine runs through all ten 90-minute lessons.

The 30 minutes

Time	Part	What happens
5 min	**Warm-up**	Million Maker: the tutor turns over digit cards one at a time; children write each digit in one of seven place-value boxes straight away and cannot move it. Closest to 5 000 000 wins. Then five quick questions on whiteboards.
8 min	**Explore**	How long is a million seconds? Pairs guess first (an hour, a day, a week, a month or a year), then work out the seconds in an hour and in a day and use them to find how many days a million seconds lasts.
5 min	**Mini-lesson**	Read 7-digit numbers in groups of three from the right; tell the value of a digit apart from how many thousands a number has. Closest to a target with digit cards: build the best number just below and the best just above, then compare the gaps. Digit riddles: use the rounding clue to fix the first digits, follow the clues in order and check every clue at the end. Before page 2: estimate by rounding to catch wrong calculator answers; brackets first, then × and ÷, then + and −; for "find all" with rounding, write the range first.
9 min	**Worksheet**	The child works on questions 1 to 4 of the Lesson 1 sheet (next pages), then checks them with the tutor.
3 min	**Exit ticket**	(a) What is the value of the digit 4 in 8 406 125? (b) Add one pair of brackets to make it true: 5 + 3 × 8 − 2 = 62

What you need

Digit cards 0 to 9 (one set per child), a 7-column place-value mat (M, HTh, TTh, Th, H, T, O), mini whiteboard, a calculator for the explore (every child needs one; a phone calculator is fine), workbook.

Online: Children cut out paper digit cards and draw a 7-box place-value mat before the lesson and hold them up to the camera. The tutor shares the slides and writes on the place-value chart. Pairs work in breakout rooms for the explore and page 1. Before the lesson, ask every parent to put a calculator (a phone calculator is fine) next to the child for the explore.

Pages in this file: this page, the child's worksheet (about two pages), then tutor notes with the answers. Do not give the tutor-notes pages to the child. This taster is enrichment for thinking skills; it does not promise exam results and is not affiliated with MOE or the Gifted Education Programme.

MILLION PUZZLES

Name _____ Group _____ Date _____

No.1

Place value. Read big numbers in groups of three from the right: 4 | 053 | 200.

a Numerals for five million, three hundred and eight thousand and forty: _____

b In 6 482 315, the digit 4 stands for _____. The ten thousands digit is _____.

c How many thousands make 3 250 000? _____

No.2

Use ALL seven digit cards, each once, to make a 7-digit number (it cannot start with 0).

0	2	3	5	6	8	9

a Greatest: _____ (b) Smallest EVEN: _____

No.3

Digit riddle. Use the clues one at a time. Then check every clue.

Clue 1. I am a 7-digit number. My digits are all different.
Clue 2. Rounded to the nearest hundred thousand, I am 7 300 000.
Clue 3. My ten thousands digit is 3 more than my hundred thousands digit.
Clue 4. My thousands digit = my ten thousands digit – my hundred thousands digit.
Clue 5. My hundreds digit is twice my thousands digit. My tens digit is 0.
Clue 6. My digits add up to 32.

M	HTh	TTh	Th	H	T	O

a Clue 2: I start 7 2 … or 7 3 … Clue 3 gives the next digit. Try both: 7 2 _____ … and 7 3 _____ …

b Clue 4 gives the next digit after that. Now check Clue 1 (all different). Which start breaks it? 7 ____ is wrong.

c Use Clues 5 and 6 to finish. I am _____

No.4

Ravi used a calculator. Estimate each answer by rounding. Which answers must be wrong?

Calculation	Ravi's answer	My estimate	Could be right / Must be wrong
(a) 4987 × 21	104 727		
(b) 3 004 512 ÷ 48	6259		
(c) 2 345 678 + 3 987 654	5 333 332		
(d) 806 × 395	318 370		

Tutor notes: Lesson 1, Million Puzzles

Numbers to 10 Million • Ages 10-12 (finishing P5)

Goal

Read, write and compare numbers up to 10 million; solve digit-card and digit-riddle puzzles with place value (testing the best number just below and just above a target); estimate to check large calculations; and use the order of operations and brackets to hit target numbers.

Skills: Place value to 10 million, testing both sides of a target, reasoning from clues in order, estimating by rounding, order of operations with brackets, making a list in order

Opening

Welcome to Math Inquirers P5! This holiday we play with puzzles that stretch your thinking. Every lesson we notice a pattern, test it and explain why.

Today our numbers grow up to 10 million. How big is a million, really? By the end you will know, and you will solve puzzles that most adults get wrong.

Warm-up (about 5 minutes)

Million Maker: the tutor turns over digit cards one at a time; children write each digit in one of seven place-value boxes straight away and cannot move it. Closest to 5 000 000 wins. Then five quick questions on whiteboards.

Draw seven boxes for millions down to ones. I turn over one digit card at a time. Write it in a box straight away. No moving it later!

The winner is the number closest to 5 000 000. Think: where does a 4 go? A 9? A 0?

Explore (about 8 minutes)

How long is a million seconds? Pairs guess first (an hour, a day, a week, a month or a year), then work out the seconds in an hour and in a day and use them to find how many days a million seconds lasts.

Predict: Before any working: how long is a million seconds? An hour, a day, a week, a month or a year? Write your guess in the corner of your whiteboard and do not change it.

Predict: Think for 30 seconds before anyone speaks, and before you touch the objects. Silent thinking. I will show the clock. Then talk with your partner.

- Stuck: how many seconds in one minute? In one hour? Multiply, do not add.
- Stuck on days: you know the seconds in one day. How many days make close to 1 000 000? Try 10 days first.
- If a pair wants more: how long is 10 million seconds? More or less than a school term?

Mini-lesson (about 5 minutes)

Read 7-digit numbers in groups of three from the right; tell the value of a digit apart from how many thousands a number has. Closest to a target with digit cards: build the best number just below and the best just above, then compare the gaps. Digit riddles: use the rounding clue to fix the first digits, follow the clues in order and check every clue at the end. Before page 2: estimate by rounding to catch wrong

calculator answers; brackets first, then × and ÷, then + and −; for "find all" with rounding, write the range first.

Worksheet answers (questions 1 to 4)

Q1	(a) 5 308 040 (b) 400 000; 8 (c) 3250
Q2	(a) 9 865 320 (b) 2 035 698
Q3	(a) 7 2 5 … and 7 3 6 … (b) 7 3 is wrong: Clue 4 gives 6 − 3 = 3, so the digit 3 would be used twice (c) 7 253 609. Rounding gives 7 2 with ten thousands digit 5 or more, or 7 3 with ten thousands digit 4 or less. "3 more" only works for 7 25: 2 + 3 = 5. Then thousands 5 − 2 = 3, hundreds 6, tens 0, and 7 + 2 + 5 + 3 + 6 + 0 = 23, so the ones digit is 32 − 23 = 9.
Q4	(a) about 100 000: could be right (b) about 60 000: must be wrong (c) about 6 300 000: must be wrong (d) about 320 000: could be right

Talk time

Siti says a 7-digit number is always bigger than a 6-digit number. Ravi says not if the 6-digit number has bigger digits, like 999 999. Who is right? Explain.

Answer: Siti is right. The smallest 7-digit number is 1 000 000 and the biggest 6-digit number is 999 999, which is 1 less. A 7-digit number has a millions digit that is at least 1, and that alone beats every 6-digit number, whatever its digits are.

Exit ticket (about 3 minutes)

(a) What is the value of the digit 4 in 8 406 125? (b) Add one pair of brackets to make it true: 5 + 3 × 8 − 2 = 62

Answer: (a) 400 000 (b) (5 + 3) × 8 − 2 = 62

Watch for these mistakes

- Leaving out zeros that hold a place, for example writing three million, forty thousand and six as 340 006 instead of 3 040 006, or mixing up the value of a digit (3000) with how many thousands a number has (4053).
- Looking only on one side of the target: for "closest to 6 000 000" trying only numbers that start with 6 and missing 5 986 320, which is closer.
- Working from left to right and ignoring the order of operations (6 + 4 × 5 read as 50 instead of 26).

Extension for fast finishers

Use the digit cards 0 to 9, each at most once, to make the 7-digit number closest to 5 000 000. (5 012 346 is 12 346 away and 4 987 653 is 12 347 away, so 5 012 346 wins by 1.)

After the taster

Tell the parent one thing their child noticed and one thing they explained well. Then share the next steps: [dates, fee and sign-up link]. The free diagnostic for the matching level helps choose between Ages 10-12 (finishing P5) and the level either side (see the placement note).